

Ex No: 2d

IMPLEMENTATION OF FORWARD CHAINING

Aim:

To implement forward Chaining.

Procedure:

Initialize initial facts and define rules as (conclusion, conditions) tuples.

Set up forward_chaining using an inferred set initialized with known facts.

Iteratively evaluate rules to check if all conditions are in inferred and add new conclusions to inferred.

Repeat the inference loop until no new facts can be derived, then check if the target goal is present.

Define the target goal, invoke forward_chaining(facts, rules, goal), and display whether the goal is achieved.

Program:

Facts: The knowledge base containing information about facts

facts = {

 'a': True,

 'b': True,

 'c': False

}

Rules: List of rules where each rule is a tuple (conclusion, conditions)

rules = [

 ('d', ['a', 'b']), # d can be concluded if a and b are true

 ('e', ['b', 'c']), # e can be concluded if b and c are true

 ('f', ['d', 'e']) # f can be concluded if d and e are true

]

Forward Chaining Function

def forward_chaining(facts, rules, goal):

```

inferred = set(facts.keys()) # Initialize inferred facts with the given facts
new_inferred = set(facts.keys()) # Start with the facts
while new_inferred:
    current_inferred = set() # Set to hold newly inferred facts
    for rule in rules:
        head, body = rule
        if head not in inferred and all(fact in inferred for fact in body):
            current_inferred.add(head)
            if current_inferred:
                inferred.update(current_inferred)
            new_inferred = current_inferred # Set new inferred facts for the next iteration
        else:
            new_inferred = set() # No new facts can be inferred

    return goal in inferred # Check if the goal is in the inferred facts

```

Example Usage

```
goal = 'f'
```

```

if forward_chaining(facts, rules, goal):
    print(f"The goal '{goal}' can be achieved.")
else:
    print(f"The goal '{goal}' cannot be achieved.")

```

Output:

The goal 'f' can be achieved.

The goal 'e' cannot be achieved.

The goal 'd' can be achieved.

```
IDLE Shell 3.11.4
File Edit Shell Debug Options Window Help
Python 3.11.4 (tags/v3.11.4:8e8a0a4, Jun 6 2023, 13:03:39) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>> ===== RESTART: C:/Users/Param/Desktop/forward_chaining.py =====
Facts:
a = True
b = True
c = False

Rules:
d :- a, b
e :- b, c
f :- d, e

Goal: f

Checking if the goal can be inferred...
Cannot infer e (because c is False).
Therefore, f cannot be inferred.

The goal 'f' cannot be achieved.

>>> |
```